

Mastering Embedded Systems Online Diploma

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FIRST TERM (FINAL PROJECT 1)

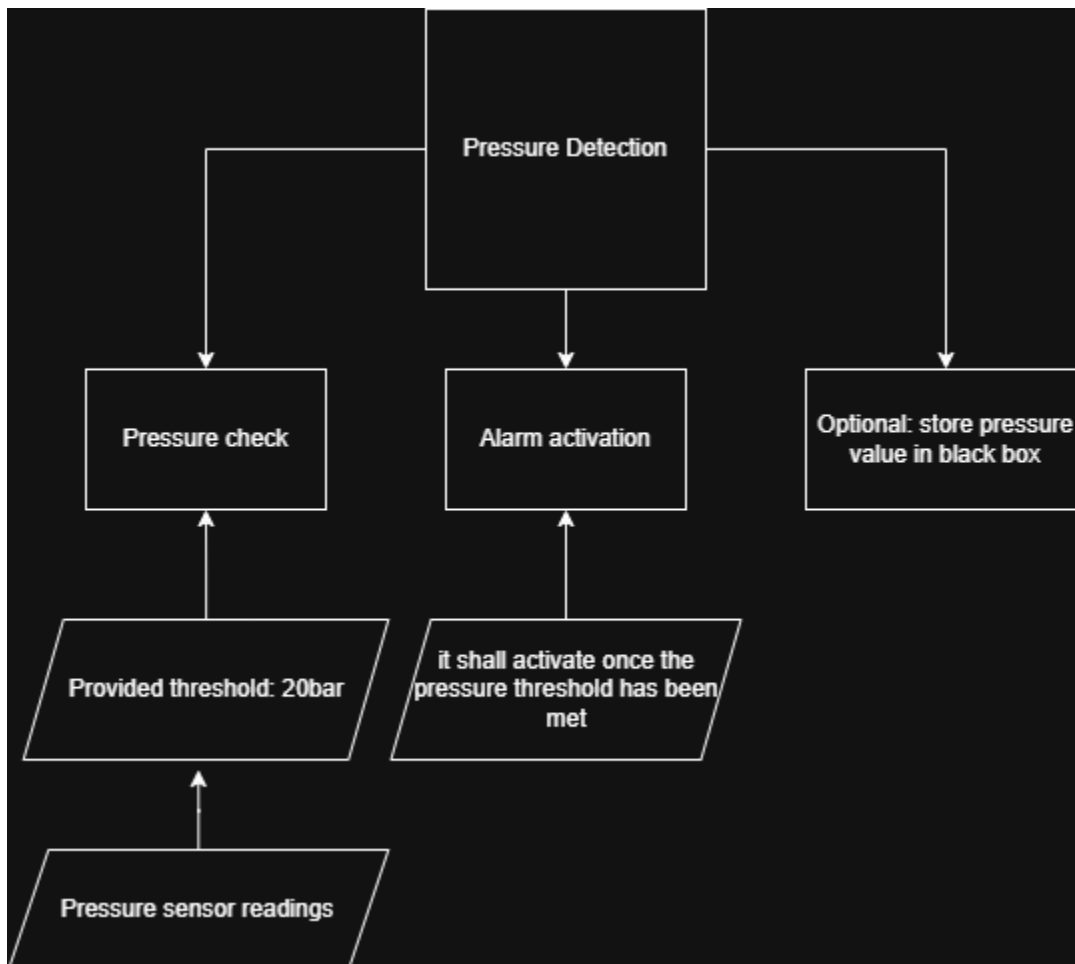
ENG. MAHMOUD ELYAMANI

MY PROFILE:

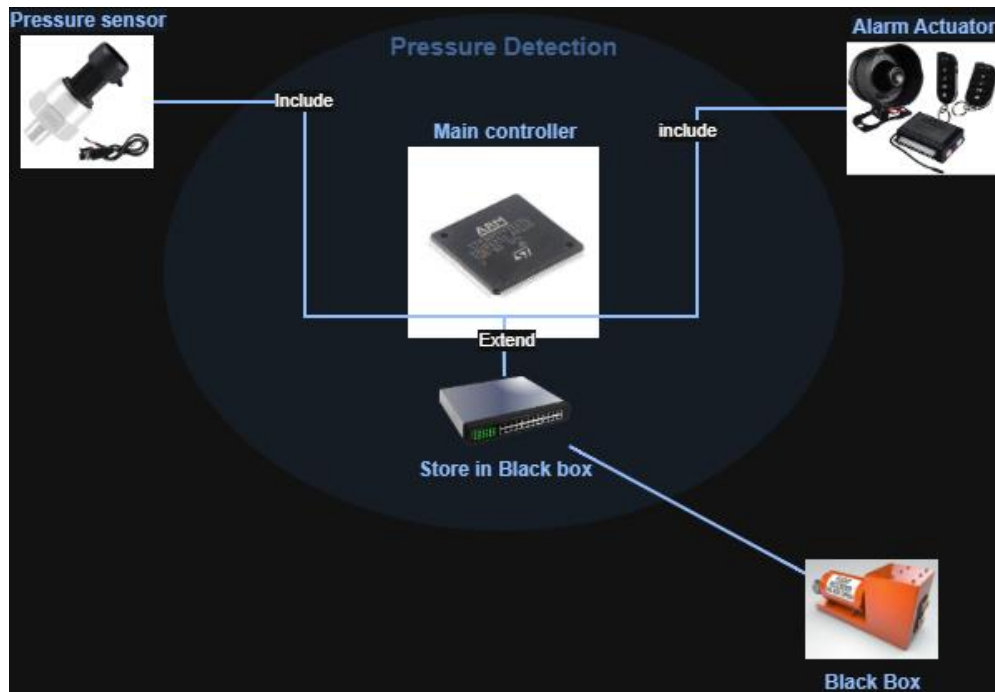
www.learn-in-depth-store.com/profile/mahmoud-elyamani-7/profile

SYSTEM REQUIREMENTS:

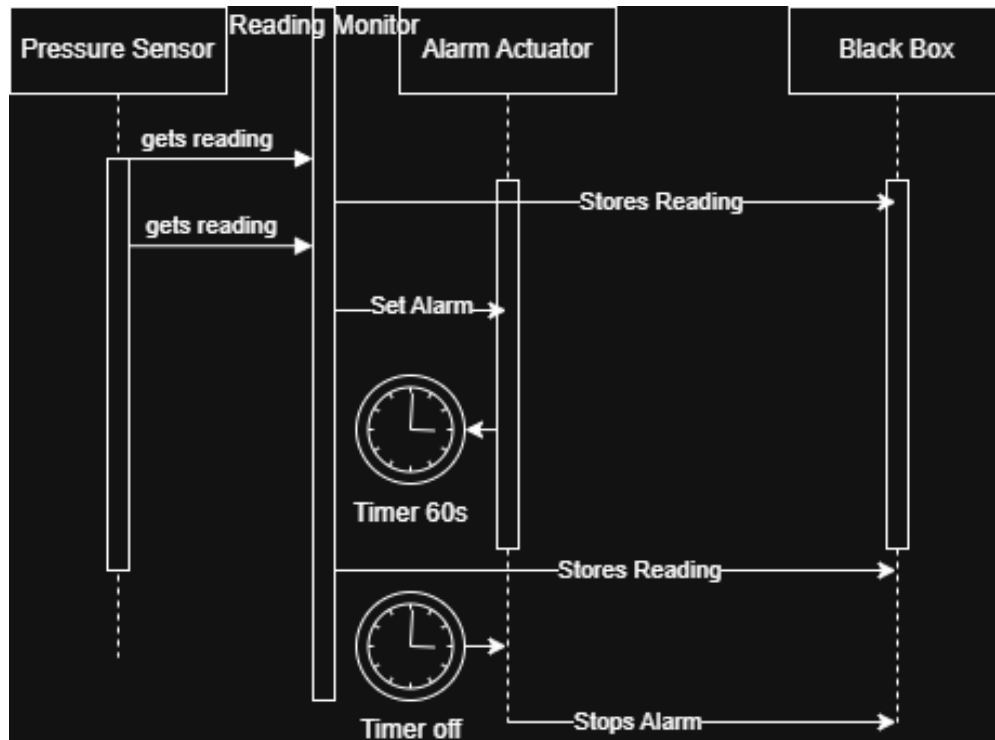
- Pressure detection firmware.
- Alarm activation firmware when pressure exceeds 20 bar.
- Activation state stays on for 60 seconds before it turns off.
- The system is assumed to be on at all time.
- The controller's turn on and turn off procedure are not taken into account.
- Pressure sensor and alarm are assumed to never fail.
- Controller's maintenance is not taken into account.



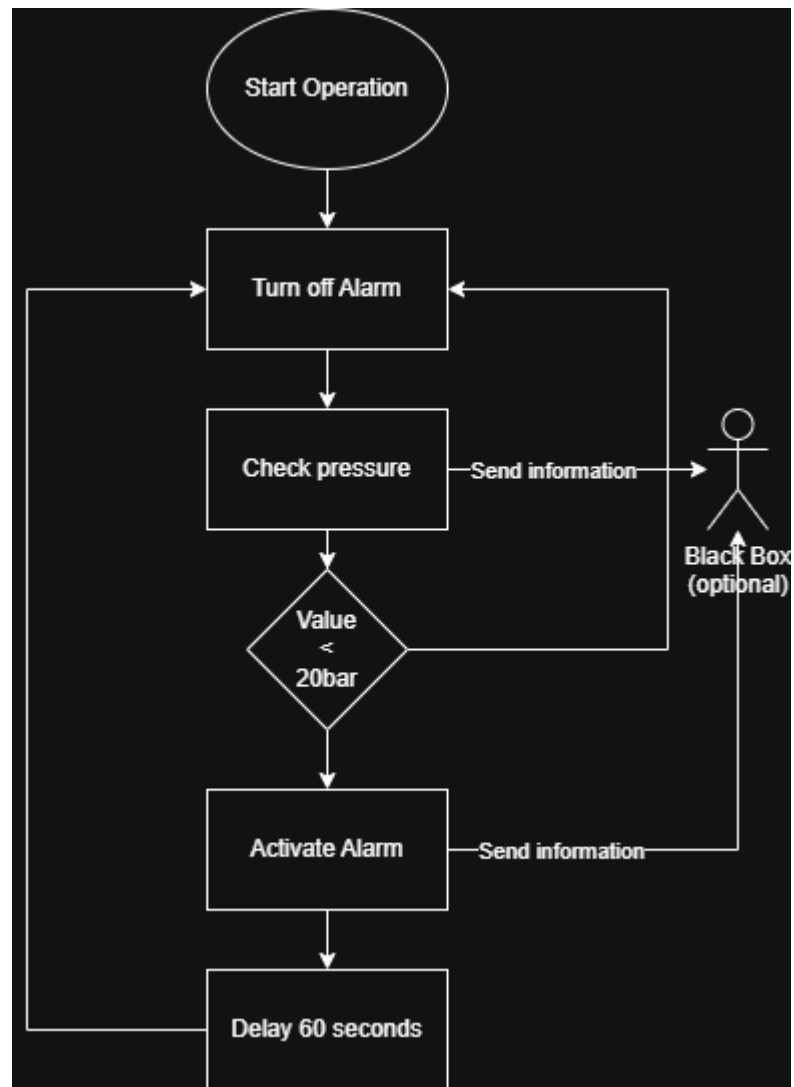
Use Case Diagram:



Sequence Diagram:



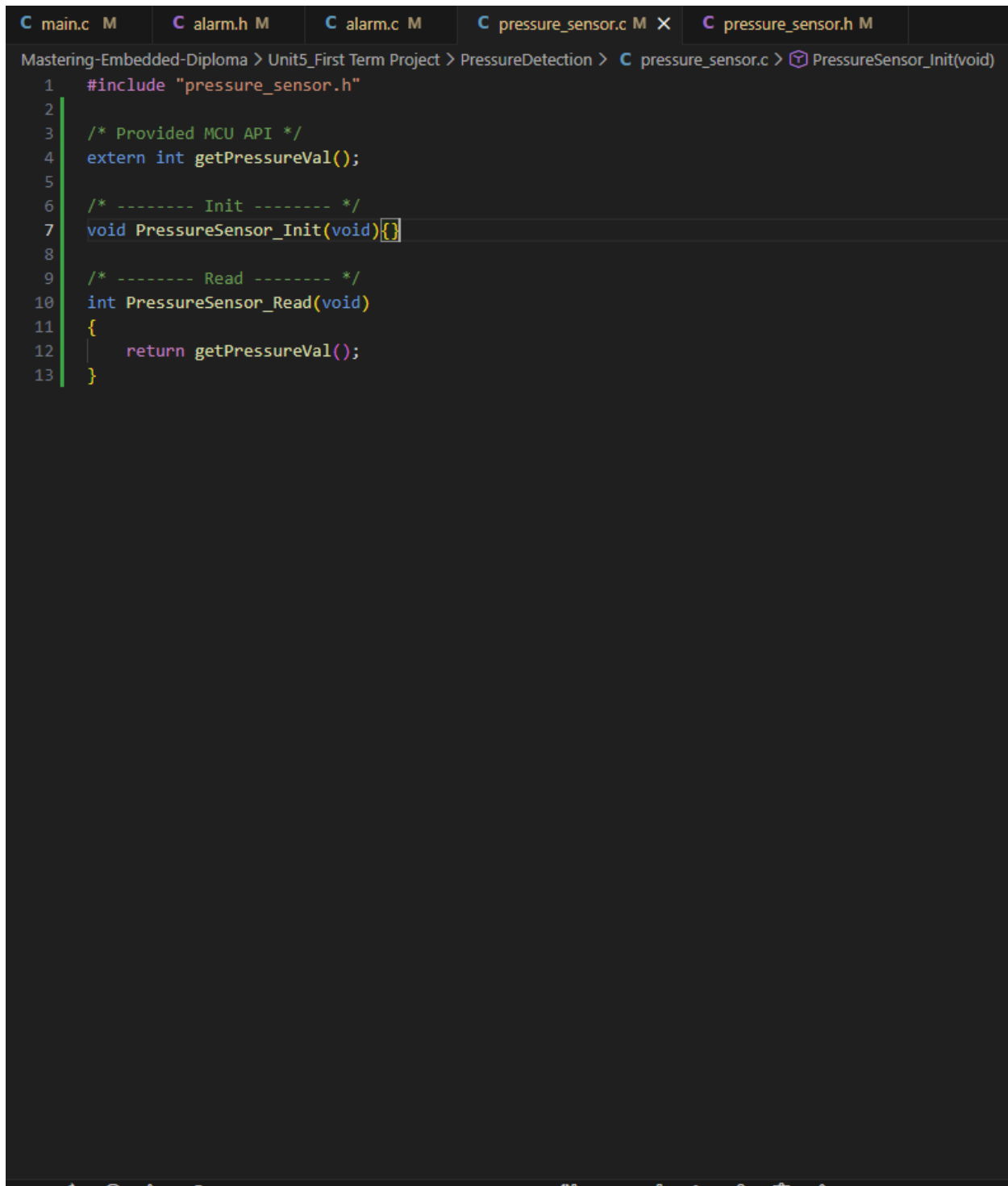
Activity Diagram:



Main.c:

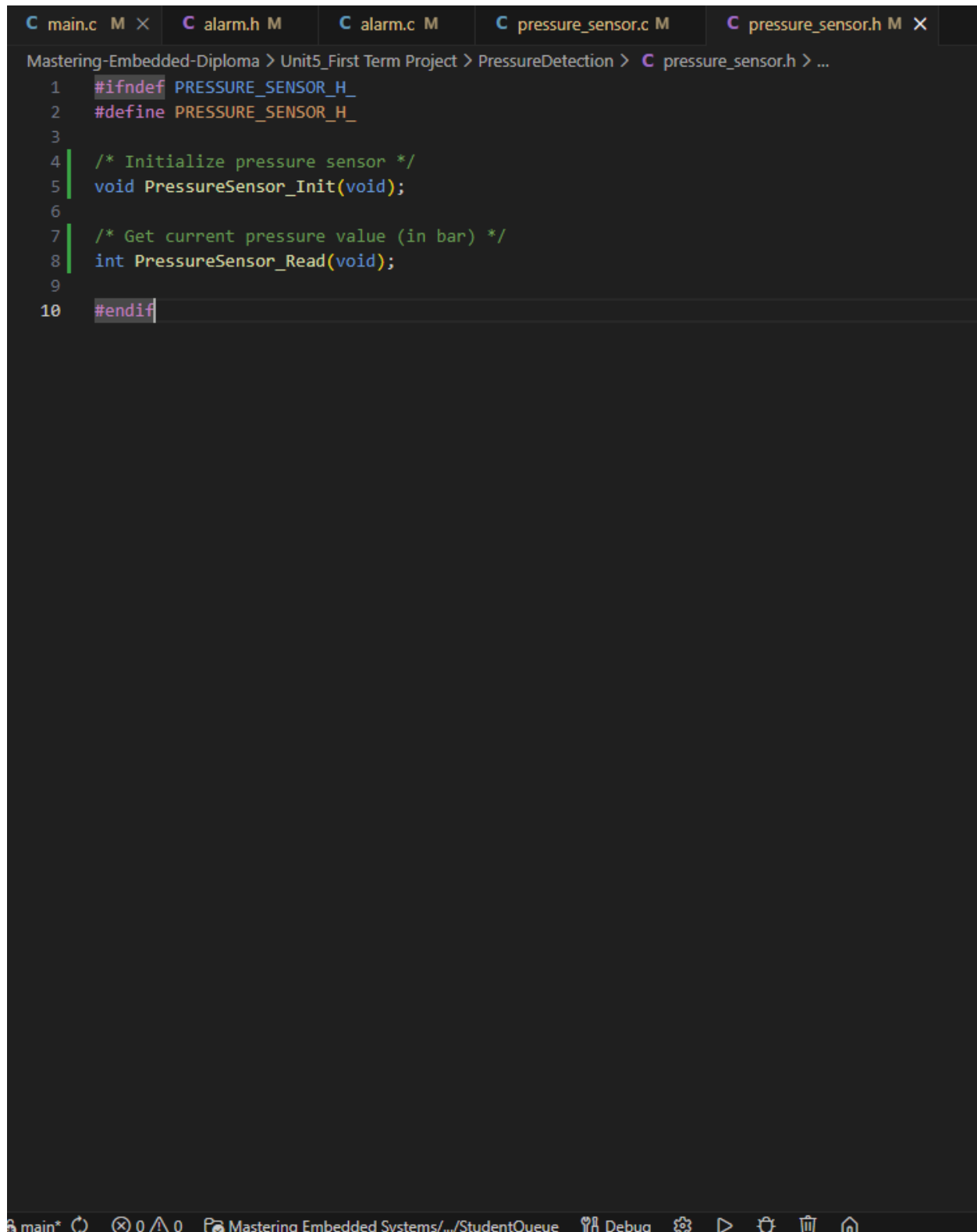
```
C main.c M X  C alarm.h M  C alarm.c M  C pressure_sensor.c M  C pressure_sensor.h M
Mastering-Embedded-Diploma > Unit5_First Term Project > PressureDetection > C main.c > main(void)
1  #include <stdint.h>
2  #include <stdio.h>
3
4  #include "driver.h"
5  #include "alarm.h"
6  #include "pressure_sensor.h"
7
8  extern void Delay(int nCount);
9  extern void GPIO_INITIALIZATION();
10
11 /* Threshold definition */
12 #define PRESSURE_THRESHOLD 20
13
14 int main(void)
15 {
16     int pressure = 0;
17
18     /* Initialize hardware */
19     GPIO_INITIALIZATION();
20     PressureSensor_Init();
21     Alarm_Init();
22
23     while (1)
24     {
25         /* Read pressure */
26         pressure = PressureSensor_Read();
27
28         /* Check threshold */
29         if (pressure >= PRESSURE_THRESHOLD)
30         {
31             Alarm_On();
32         }
33         else
34         {
35             Alarm_Off();
36         }
37
38         /* 60 seconds delay */
39         Delay(60000);
40     }
41
42     return 0;
43 }
```

Pressure_Sensor.c:



```
1  #include "pressure_sensor.h"
2
3  /* Provided MCU API */
4  extern int getPressureVal();
5
6  /* ----- Init ----- */
7  void PressureSensor_Init(void){}
8
9  /* ----- Read ----- */
10 int PressureSensor_Read(void)
11 {
12     return getPressureVal();
13 }
```

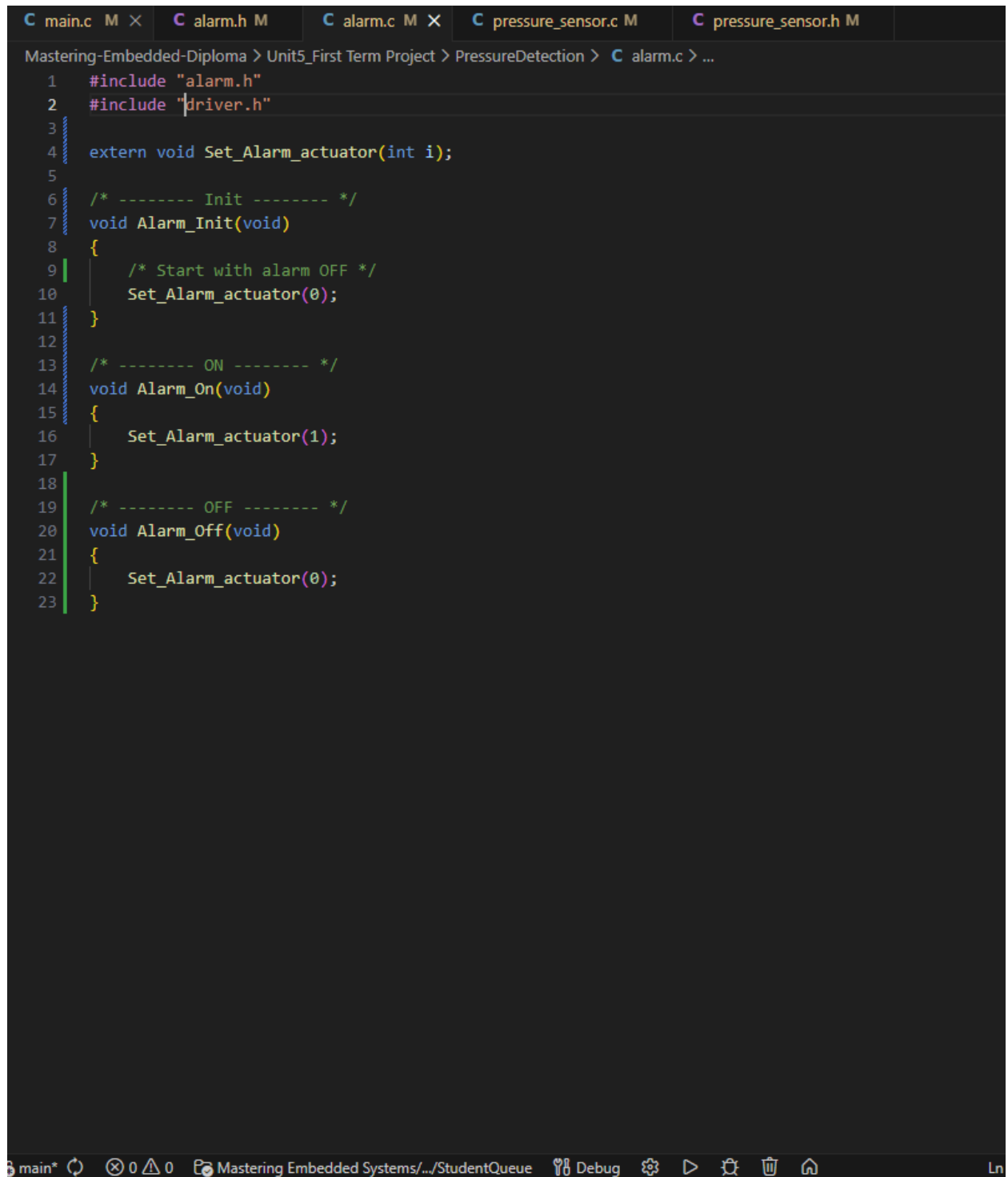
Pressure_Sensor.h:



```
Mastering-Embedded-Diploma > Unit5_First Term Project > PressureDetection > C pressure_sensor.h > ...
1  #ifndef PRESSURE_SENSOR_H_
2  #define PRESSURE_SENSOR_H_
3
4  /* Initialize pressure sensor */
5  void PressureSensor_Init(void);
6
7  /* Get current pressure value (in bar) */
8  int PressureSensor_Read(void);
9
10 #endif
```

main* 0/0 Mastering Embedded Systems/.../StudentQueue Debug

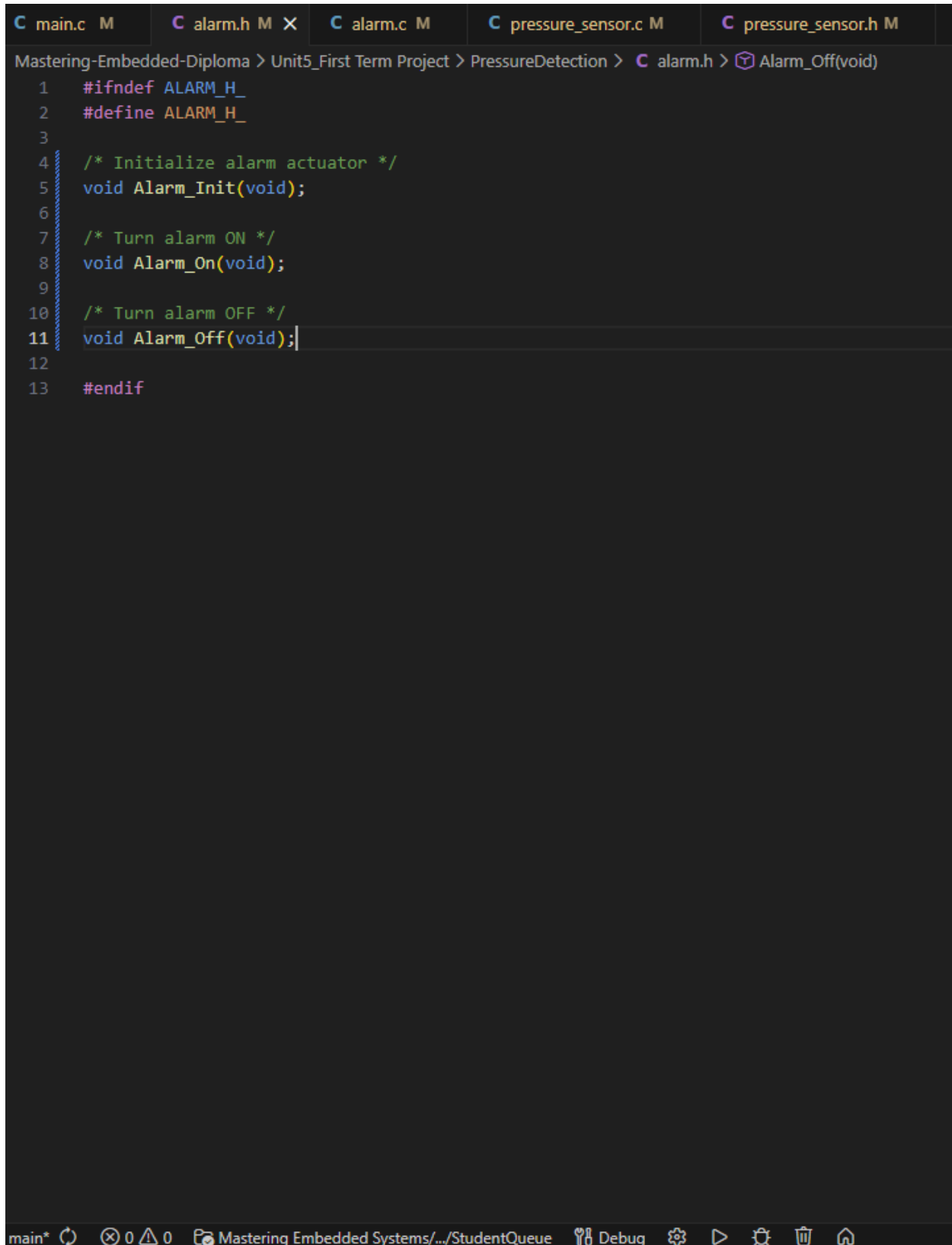
Alarm.c:



```
1  #include "alarm.h"
2  #include "driver.h"
3
4  extern void Set_Alarm_actuator(int i);
5
6  /* ----- Init ----- */
7  void Alarm_Init(void)
8  {
9      /* Start with alarm OFF */
10     Set_Alarm_actuator(0);
11 }
12
13 /* ----- ON ----- */
14 void Alarm_On(void)
15 {
16     Set_Alarm_actuator(1);
17 }
18
19 /* ----- OFF ----- */
20 void Alarm_Off(void)
21 {
22     Set_Alarm_actuator(0);
23 }
```

main* 0 0 Mastering Embedded Systems/.../StudentQueue Debug

Alarm.h:



The image shows a code editor window with a dark theme. At the top, there are four tabs: 'main.c M', 'alarm.h M X', 'alarm.c M', and 'pressure_sensor.c M'. The 'alarm.h' tab is active. Below the tabs, the breadcrumb path reads 'Mastering-Embedded-Diploma > Unit5_First Term Project > PressureDetection > alarm.h > Alarm_Off(void)'. The main editor area contains the following C code:

```
1  #ifndef ALARM_H_
2  #define ALARM_H_
3
4  /* Initialize alarm actuator */
5  void Alarm_Init(void);
6
7  /* Turn alarm ON */
8  void Alarm_On(void);
9
10 /* Turn alarm OFF */
11 void Alarm_Off(void);
12
13 #endif
```

At the bottom of the editor, there is a status bar with the following information: 'main*' followed by a refresh icon, '0 0' with a warning icon, the file path 'Mastering Embedded Systems/.../StudentQueue', and several icons for 'Debug', settings, play, stop, and other IDE functions.

Alarm.o headers&symbols:

```
MINGW64:/d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_First Term Pr...
mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ arm-none-eabi-objdump.exe -h alarm.o

alarm.o:      file format elf32-littlearm

Sections:
Idx Name          Size      VMA       LMA       File off  Algn
  0 .text          00000030 00000000 00000000 00000034 2**2
    CONTENTS, ALLOC, LOAD, RELOC, READONLY, CODE
  1 .data          00000000 00000000 00000000 00000064 2**0
    CONTENTS, ALLOC, LOAD, DATA
  2 .bss           00000000 00000000 00000000 00000064 2**0
    ALLOC
  3 .debug_info    000000bc 00000000 00000000 00000064 2**0
    CONTENTS, RELOC, READONLY, DEBUGGING
  4 .debug_abbrev  00000042 00000000 00000000 00000120 2**0
    CONTENTS, READONLY, DEBUGGING
  5 .debug_loc     00000084 00000000 00000000 00000162 2**0
    CONTENTS, READONLY, DEBUGGING
  6 .debug_aranges 00000020 00000000 00000000 000001e6 2**0
    CONTENTS, RELOC, READONLY, DEBUGGING
  7 .debug_line    0000003d 00000000 00000000 00000206 2**0
    CONTENTS, RELOC, READONLY, DEBUGGING
  8 .debug_str     00000128 00000000 00000000 00000243 2**0
    CONTENTS, READONLY, DEBUGGING
  9 .comment       00000012 00000000 00000000 0000036b 2**0
    CONTENTS, READONLY
10 .ARM.attributes 00000033 00000000 00000000 0000037d 2**0
    CONTENTS, READONLY
11 .debug_frame   00000064 00000000 00000000 000003b0 2**2
    CONTENTS, RELOC, READONLY, DEBUGGING

mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ arm-none-eabi-nm.exe alarm.o
00000000 T Alarm_Init
00000020 T Alarm_Off
00000010 T Alarm_On
          U Set_Alarm_actuator

mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ |
```

Pressure_Sensor.o headers&symbols:

```
MINGW64:/d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_First Term Pr...
U Set_Alarm_actuator

mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ arm-none-eabi-objdump.exe -h pressure_sensor.o

pressure_sensor.o:      file format elf32-littlearm

Sections:
Idx Name              Size      VMA       LMA       File off  Algn
  0 .text              0000001c  00000000  00000000  00000034  2**2
                        CONTENTS, ALLOC, LOAD, RELOC, READONLY, CODE
  1 .data              00000000  00000000  00000000  00000050  2**0
                        CONTENTS, ALLOC, LOAD, DATA
  2 .bss               00000000  00000000  00000000  00000050  2**0
                        ALLOC
  3 .debug_info        0000005d  00000000  00000000  00000050  2**0
                        CONTENTS, RELOC, READONLY, DEBUGGING
  4 .debug_abbrev      00000051  00000000  00000000  000000ad  2**0
                        CONTENTS, READONLY, DEBUGGING
  5 .debug_loc         00000058  00000000  00000000  000000fe  2**0
                        CONTENTS, READONLY, DEBUGGING
  6 .debug_aranges     00000020  00000000  00000000  00000156  2**0
                        CONTENTS, RELOC, READONLY, DEBUGGING
  7 .debug_line        00000043  00000000  00000000  00000176  2**0
                        CONTENTS, RELOC, READONLY, DEBUGGING
  8 .debug_str         000000aa  00000000  00000000  000001b9  2**0
                        CONTENTS, READONLY, DEBUGGING
  9 .comment           00000012  00000000  00000000  00000263  2**0
                        CONTENTS, READONLY
10 .ARM.attributes    00000033  00000000  00000000  00000275  2**0
                        CONTENTS, READONLY
11 .debug_frame       00000044  00000000  00000000  000002a8  2**2
                        CONTENTS, RELOC, READONLY, DEBUGGING

mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ arm-none-eabi-nm.exe pressure_sensor.o
                 U getPressureVal
00000000 T PressureSensor_Init
0000000c T PressureSensor_Read

mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ |
```

Main.o Headers&symbols:

```
MINGW64:/d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_First Term Pr...
mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ arm-none-eabi-objdump.exe -h main.o

main.o:      file format elf32-littlearm

Sections:
Idx Name          Size      VMA           LMA           File off  Algn
  0 .text          00000038  00000000  00000000  00000034  2**2
    CONTENTS, ALLOC, LOAD, RELOC, READONLY, CODE
  1 .data           00000000  00000000  00000000  0000006c  2**0
    CONTENTS, ALLOC, LOAD, DATA
  2 .bss            00000000  00000000  00000000  0000006c  2**0
    ALLOC
  3 .debug_info     000000a3  00000000  00000000  0000006c  2**0
    CONTENTS, RELOC, READONLY, DEBUGGING
  4 .debug_abbrev   00000053  00000000  00000000  0000010f  2**0
    CONTENTS, READONLY, DEBUGGING
  5 .debug_loc      00000038  00000000  00000000  00000162  2**0
    CONTENTS, READONLY, DEBUGGING
  6 .debug_aranges  00000020  00000000  00000000  0000019a  2**0
    CONTENTS, RELOC, READONLY, DEBUGGING
  7 .debug_line     00000040  00000000  00000000  000001ba  2**0
    CONTENTS, RELOC, READONLY, DEBUGGING
  8 .debug_str      00000117  00000000  00000000  000001fa  2**0
    CONTENTS, READONLY, DEBUGGING
  9 .comment        00000012  00000000  00000000  00000311  2**0
    CONTENTS, READONLY
10 .ARM.attributes 00000033  00000000  00000000  00000323  2**0
    CONTENTS, READONLY
11 .debug_frame     00000030  00000000  00000000  00000358  2**2
    CONTENTS, RELOC, READONLY, DEBUGGING

mah.elyamany@Technical2 MINGW64 /d/Mastering Embedded Systems/Mastering-Embedded-Diploma/Unit5_
First Term Project/PressureDetection (main)
$ arm-none-eabi-nm.exe main.o
                 U Alarm_Init
                 U Alarm_Off
                 U Alarm_On
                 U Delay
                 U GPIO_INITIALIZATION
00000000 T main
                 U PressureSensor_Init
                 U PressureSensor_Read
```

Mapfile:

```

1 |
2 | Memory Configuration
3 |
4 | Name          Origin          Length          Attributes
5 | FLASH         0x08000000      0x00020000      xr
6 | SRAM          0x20000000      0x00050000      xrw
7 | *default*     0x00000000      0xffffffff
8 |
9 | Linker script and memory map
10 |
11 |
12 | .TEXT          0x08000000      0x294
13 | *(.VECTORS*)
14 | .VECTORS       0x08000000      0x1c startup.o
15 |                0x08000000      vectors
16 | *(.text*)
17 | .text          0x0800001c      0x30 alarm.o
18 |                0x0800001c      Alarm_Init
19 |                0x0800002c      Alarm_On
20 |                0x0800003c      Alarm_Off
21 | .text          0x0800004c      0x10c driver.o
22 |                0x0800004c      Delay
23 |                0x08000070      getPressureVal
24 |                0x08000088      Set_Alarm_actuator
25 |                0x080000d8      GPIO_INITIALIZATION
26 | .text          0x08000158      0x38 main.o
27 |                0x08000158      main
28 | .text          0x08000190      0x1c pressure_sensor.o
29 |                0x08000190      PressureSensor_Init
30 |                0x0800019c      PressureSensor_Read
31 | .text          0x080001ac      0xe8 startup.o
32 |                0x080001ac      RESET
33 |                0x08000288      BUS_FAULT
34 |                0x08000288      DEFAULT_HANDLER
35 |                0x08000288      NMI
36 |                0x08000288      USGE_FAULT
37 |                0x08000288      MEM_MANAGE
38 |                0x08000288      HARD_FAULT
39 | *(.rodata)
40 |                0x08000294      E_TEXT = .
41 |
42 | .glue_7        0x08000294      0x0
43 | .glue_7        0x00000000      0x0 linker stubs
44 |
45 | .glue_7t       0x08000294      0x0
46 | .glue_7t       0x00000000      0x0 linker stubs

```

C	main.c	M	C	alarm.h	M	C	alarm.c	M	C	pressure_sensor.c	M
Mastering-Embedded-Diploma > Unit5_First Term Project > PressureDetection > Map_fi											
45	.glue_7t		0x08000294			0x0					
46	.glue_7t		0x00000000			0x0	linker	stubs			
47											
48	.vfp11_veneer		0x08000294			0x0					
49	.vfp11_veneer		0x00000000			0x0	linker	stubs			
50											
51	.v4_bx		0x08000294			0x0					
52	.v4_bx		0x00000000			0x0	linker	stubs			
53											
54	.iplt		0x08000294			0x0					
55	.iplt		0x00000000			0x0	alarm.o				
56											
57	.igot.plt		0x08000294			0x0					
58	.igot.plt		0x00000000			0x0	alarm.o				
59											
60	.rel.dyn		0x08000294			0x0					
61	.rel.iplt		0x00000000			0x0	alarm.o				
62											
63	.DATA		0x20000000			0x0	load address	0x08000294			
64			0x20000000				S_DATA = .				
65	*(.data*)										
66	.data		0x20000000			0x0	alarm.o				
67	.data		0x20000000			0x0	driver.o				
68	.data		0x20000000			0x0	main.o				
69	.data		0x20000000			0x0	pressure_sensor.o				
70	.data		0x20000000			0x0	startup.o				
71			0x20000000				E_DATA = .				
72											
73	.BSS		0x20000000			0x3ec	load address	0x08000294			
74			0x20000000				S_BSS = .				
75	*(.bss*)										
76	.bss		0x20000000			0x0	alarm.o				
77	.bss		0x20000000			0x0	driver.o				
78	.bss		0x20000000			0x0	main.o				
79	.bss		0x20000000			0x0	pressure_sensor.o				
80	.bss		0x20000000			0x4	startup.o				
81			0x20000000				i				
82			0x20000004				E_BSS = .				
83			0x200003ec				. = (. + 0x3e8)				
84	*fill*		0x20000004			0x3e8					
85			0x200003ec				STACK_TOP = .				
86	LOAD	alarm.o									
87	LOAD	driver.o									
88	LOAD	main.o									
89	LOAD	pressure_sensor.o									
90	LOAD	startup.o									

```

91  OUTPUT(PressureSystem.elf elf32-littlearm)
92
93  .debug_info      0x00000000      0x42b
94  .debug_info      0x00000000      0xbc alarm.o
95  .debug_info      0x000000bc      0x103 driver.o
96  .debug_info      0x000001bf      0xa3 main.o
97  .debug_info      0x00000262      0x5d pressure_sensor.o
98  .debug_info      0x000002bf      0x16c startup.o
99
100 .debug_abbrev     0x00000000      0x247
101 .debug_abbrev     0x00000000      0x42 alarm.o
102 .debug_abbrev     0x00000042      0x9d driver.o
103 .debug_abbrev     0x000000df      0x53 main.o
104 .debug_abbrev     0x00000132      0x51 pressure_sensor.o
105 .debug_abbrev     0x00000183      0xc4 startup.o
106
107 .debug_loc        0x00000000      0x240
108 .debug_loc        0x00000000      0x84 alarm.o
109 .debug_loc        0x00000084      0xc8 driver.o
110 .debug_loc        0x0000014c      0x38 main.o
111 .debug_loc        0x00000184      0x58 pressure_sensor.o
112 .debug_loc        0x000001dc      0x64 startup.o
113
114 .debug_aranges    0x00000000      0xa0
115 .debug_aranges    0x00000000      0x20 alarm.o
116 .debug_aranges    0x00000020      0x20 driver.o
117 .debug_aranges    0x00000040      0x20 main.o
118 .debug_aranges    0x00000060      0x20 pressure_sensor.o
119 .debug_aranges    0x00000080      0x20 startup.o
120
121 .debug_line        0x00000000      0x20e
122 .debug_line        0x00000000      0x3d alarm.o
123 .debug_line        0x0000003d      0x9c driver.o
124 .debug_line        0x000000d9      0x40 main.o
125 .debug_line        0x00000119      0x43 pressure_sensor.o
126 .debug_line        0x0000015c      0xb2 startup.o
127
128 .debug_str         0x00000000      0x206
129 .debug_str         0x00000000      0xef alarm.o
130
131 (size before relaxing)

```


C	main.c	M	×	C	alarm.h	M	C	alarm.c	M	C	pressure_sensor.c	M
Mastering-Embedded-Diploma > Unit5_First Term Project > PressureDetection >  Map_fi												
126	.debug_line			0x00000000			0x20e					
127	.debug_line			0x0000003d			0x3c					driver.o
129	.debug_line			0x000000d9			0x40					main.o
130	.debug_line			0x00000119			0x43					pressure_sensor.o
131	.debug_line			0x0000015c			0xb2					startup.o
132												
133	.debug_str			0x00000000			0x206					
134	.debug_str			0x00000000			0xef					alarm.o
135							0x128					(size before relaxing)
136	.debug_str			0x000000ef			0x55					driver.o
137							0x157					(size before relaxing)
138	.debug_str			0x00000144			0x15					main.o
139							0x117					(size before relaxing)
140	.debug_str			0x00000159			0x3a					pressure_sensor.o
141							0xaa					(size before relaxing)
142	.debug_str			0x00000193			0x73					startup.o
143							0x179					(size before relaxing)
144												
145	.comment			0x00000000			0x11					
146	.comment			0x00000000			0x11					alarm.o
147							0x12					(size before relaxing)
148	.comment			0x00000000			0x12					driver.o
149	.comment			0x00000000			0x12					main.o
150	.comment			0x00000000			0x12					pressure_sensor.o
151	.comment			0x00000000			0x12					startup.o
152												
153	.ARM.attributes											
154				0x00000000			0x33					
155	.ARM.attributes											
156				0x00000000			0x33					alarm.o
157	.ARM.attributes											
158				0x00000033			0x33					driver.o
159	.ARM.attributes											
160				0x00000066			0x33					main.o
161	.ARM.attributes											
162				0x00000099			0x33					pressure_sensor.o
163	.ARM.attributes											
164				0x000000cc			0x33					startup.o
165												
166	.debug_frame			0x00000000			0x19c					
167	.debug_frame			0x00000000			0x64					alarm.o
168	.debug_frame			0x00000064			0x78					driver.o
169	.debug_frame			0x000000dc			0x30					main.o
170	.debug_frame			0x0000010c			0x44					pressure_sensor.o
171	.debug_frame			0x00000150			0x4c					startup.o
172												

Simulation on Proteus:

